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Lab	3
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Lab Report:

This laboratory exercise provided hands-on experience with a professional multi-function network cable tester designed to verify the continuity, wiring configuration, and termination quality of various network cables, including RJ-45 (Ethernet), RJ-11 (telephone), USB, and BNC (coaxial) types. By systematically testing cables and interpreting the device's LED indicators, we identified common cabling faults such as shorts, open circuits, crossed pairs, and improper terminations. The experiment underscored the critical role of cable testing in ensuring reliable data transmission, minimizing network downtime, and maintaining high-performance communication infrastructure in modern computer networks.

1. Introduction

In the field of computer networks, the physical layer forms the foundation of all data communication. Even the most advanced networking protocols and devices cannot compensate for faults in the underlying cabling infrastructure. Poorly crimped, damaged, or incorrectly wired cables can lead to packet loss, reduced bandwidth, intermittent connectivity, and complete network failures. According to industry standards, regular cable testing and certification are essential practices during network installation, maintenance, and troubleshooting. This third laboratory session in the Computer Networks course focused on the practical use of a handheld network cable tester. The primary goal was to develop skills in diagnosing cable related issues using specialized testing equipment. Understanding how to quickly and accurately verify cable integrity is a fundamental competency for network administrators, technicians, and engineers responsible for deploying and maintaining local area networks (LANs), data centers, and telecommunication systems.

2. Objectives

The objectives of this lab were as follows:

- To understand the working principle and features of a multi-purpose network cable tester.
- To learn how to test different types of network cables (RJ-45, RJ-11, USB, and BNC) for continuity and wiring correctness.
- To identify and interpret common cable faults, including shorts, open connections, non-parallel (crossed) wires, and termination problems.

- To appreciate the importance of cable testing in preventing network performance degradation and ensuring compliance with cabling standards (such as TIA/EIA-568 for structured cabling).

3. Theory and Background

Network cables serve as the medium for transmitting electrical or optical signals between devices. Common cable types include:

- **Unshielded Twisted Pair (UTP)** cables terminated with RJ-45 connectors (used for Ethernet networks supporting 10/100/1000 Mbps and higher speeds).
- **RJ-11** connectors (typically for telephone lines).
- **USB cables** (for peripheral connectivity).
- **BNC coaxial cables** (used in older video or legacy network installations).

A network cable tester works by sending test signals through each conductor (wire) of the cable and measuring continuity, resistance, and signal integrity at the receiving end. The device compares the received signals against expected patterns to detect:

- **Continuity:** Whether each wire is properly connected end-to-end.
- **Shorts:** Unintended contact between two or more wires.
- **Opens:** Broken or disconnected wires.
- **Miswiring/Crosses:** Incorrect pin assignments or crossed pairs (non-parallel wiring).
- **Termination Issues:** Especially important for coaxial cables (e.g., correct 50Ω or 25Ω termination vs. no terminator).

The tester used in this lab features separate testing sections for RJ-45/RJ-11/USB and BNC cables, along with visual LED indicators for pin status (S, C1–C8), fault types, and battery condition. Proper testing helps ensure cables meet performance requirements for reliable, high speed data transmission.

4. Apparatus and Materials

- Professional handheld network cable tester (multi-function device supporting RJ-45, RJ 11, USB, and BNC).
- Various network cables for testing (straight-through RJ-45 Ethernet cables, patch cords, and sample BNC cables).
- Notebook and pen for recording observations.

5. Procedure

1. Device Familiarization: We examined the cable tester's layout, including the top row of LEDs for pin/wire condition, status indicators for faults (SHORT, CONNECTED, NON- Practical Analysis and Testing of Network Cables Using a Multi-Function Cable Tester PARALLEL, NO CONNECTION), BNC-specific termination indicators, and battery status (GOOD/LOW).

2. Power On and Self-Check: The tester was powered on, and battery status was verified (green LED indicated "BATTERY GOOD").

3. Cable Connection: Different cables were connected one by one to the appropriate ports on the tester (RJ-45 into the main port, BNC into the coaxial port).

4. Testing Process: The tester automatically cycled through each conductor. We observed:

- o Sequential lighting of orange LEDs (indicating proper pin continuity).
- o Activation of status LEDs (e.g., blue "CONNECTED" LED for good cables).
- o Any fault indications (red "SHORT", crossed symbols for non-parallel wiring, or "NO CONNECTION").

5. Fault Simulation and Diagnosis: Where possible, we tested known good cables versus intentionally faulty ones to observe different LED patterns.

6. Documentation: Results were noted, and the device in operation was photographed for inclusion in the report.

6. Observations and Results

During the practical session, multiple RJ-45 Ethernet cables were tested. Most cables showed proper sequential illumination of the pin LEDs along with a steady blue "CONNECTED" indicator, confirming correct wiring and continuity.

The provided image captures the cable tester in active use with a cable connected (visible on the right side). The top row of orange LEDs is partially lit, indicating ongoing pin testing, while the blue LED under "CONNECTED" is illuminated, and the battery status shows "BATTERY GOOD." This visual evidence demonstrates the device actively verifying cable integrity.



7. Discussion

The lab demonstrated that even minor cabling errors can severely impact network performance. For instance, a single crossed pair in a Gigabit Ethernet cable can cause significant packet errors and speed reduction. The cable tester proved to be an efficient, portable tool for rapid diagnosis without needing complex equipment like a full cable certifier. This practical reinforced theoretical concepts from lectures on the OSI model's Physical layer and structured cabling standards. In real-world scenarios, such testing is performed during initial installation, after moves/adds/changes, and as part of routine maintenance to prevent costly downtime.

8. Conclusion

This laboratory session successfully achieved its objectives by providing practical exposure to professional cable testing equipment. We gained valuable skills in identifying

and interpreting cable faults, which are essential for building and maintaining robust computer networks. The exercise highlighted that proactive cable verification is a key best practice in network engineering, contributing directly to network reliability, performance optimization, and reduced troubleshooting time. Future labs could extend this work by incorporating cable creation (crimping straight-through and crossover cables) followed by immediate testing, or by exploring advanced testers with length measurement and certification capabilities.